The Optimal Strictness of Up-or-Out Contracts*

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Abstract

Up-or-out contracts place junior workers on fixed-term probation, evaluate them against a standard, and then either promote them to higher positions or end the relationship. Organisations share this blueprint, but their designs vary widely. The promotion threshold is meant to retain agents with a sound record at the end of probation. A higher threshold selects better agents but at the cost of dismissing those who are capable enough to take on senior responsibility. The optimal threshold balances these two effects. Once probation length enters the design problem as well, the organisation must weigh both against the cost of keeping evaluation open for longer.

This paper studies the design of up-or-out contracts from the ability-revealing side. The agent picks a project from a feasible pool. Projects differ in the variance of their outcomes and hence in their informativeness about the agent's ability. The agent maximises the promotion probability by selecting the project with the optimal ability disclosure.

In the single-period baseline, an informative equilibrium exists when the feasible project menu is wide enough. The organisation sets a strict enough threshold to make the agent prefer the most informative project. In the multi-period extension, the agent's optimal early-period precision is interior. Two forces shape it. The recovery force keeps open the possibility of a strong recovery in the final stage if the early project goes badly. The safety force maintains a higher promotion probability if the early project goes well. On the organisation side, threshold strictness and probation length are substitutes. Short probation calls for strict standards. Long probation calls for lenient ones. High cost of extending probation favours the strict-short regime; low cost favours the lenient-long one. A quota extension shows that promoting a fixed top-k of N cohort members is asymptotically equivalent to committing to a threshold.

JEL codes: D82, D86, J33, M51.
Keywords: up-or-out contracts, probation, career concerns, project choice, information design.

1 Introduction

Organisations follow sophisticated procedures in production. Rules are set to assign more responsibility to competent people. In practice, up-or-out contracts help law firms select sound lawyers to handle more difficult cases; tenure attracts and retains bright minds in universities.

Up-or-out contracts place junior workers on fixed-term probation, evaluate them against a standard, and then either promote them to higher positions or end the relationship. While organisations share the blueprint of up-or-out, their design varies widely. The effect of contract design is multidimensional; Huang et al. (2025) document that up-or-out contracts raise productivity through three channels: the recruitment of high-potential researchers, heightened performance pressure from employment risk, and enhanced incentives such as higher salaries and research

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funding. This paper focuses on the ability-revealing side and explains how the optimal design of an up-or-out contract contributes to a better estimation of the agent's ability.

The promotion threshold is meant to retain agents with a sound record at the end of probation. A higher threshold selects better agents but at the cost of dismissing those who are capable enough to take on senior responsibility. The optimal threshold balances those two effects; once probation length enters the design problem as well, the organisation must weigh both against the cost of keeping evaluation open for longer.

On the information side, the promotion decision reflects whether the posterior estimate of the agent's ability exceeds a threshold. The agent selects a project from a feasible pool. Projects differ in the variance of their outcomes and hence in their informativeness about the agent's ability. The agent maximises the promotion probability by selecting the project with the optimal ability disclosure.

In the baseline model, we characterise an *informative equilibrium*, in which the organisation sets a sufficiently strict threshold to make the agent prefer the project whose outcome is most informative about the agent's ability (Proposition 5.1). The existence rests on the range of the feasible project menu. The organisation only sets a higher threshold once it can extract enough informational advantage to justify the cost of sub-optimal screening.

Extending the baseline to a multi-period setup uncovers a trade-off. When the agent can adapt to past outcomes, she prefers to reveal limited information about her ability in the early stages. Such a strategy is shaped by two forces. The recovery force keeps open the possibility of a strong recovery in the final stage if the early project goes badly. The safety force maintains a higher promotion probability if the early project goes well. Marginally raising first-period precision strengthens the safety force and weakens the recovery force; this trade-off allows us to pin down the optimum by a marginal cost equals marginal revenue condition. On the organisation side, threshold strictness and probation length are substitutes. We establish this under constant-precision bounds on the agent's play, since her general multi-period best response lacks a closed form. Short probation calls for strict standards; long probation calls for lenient ones. The cost of extending the screening process distinguishes which regime is preferred: high cost favours the strict-short regime, low cost favours the lenient-long one.

The rest of the paper proceeds as follows. Section 2 sets out the single-period model and characterises the informative equilibrium. Section 6 adds multi-period probation, derives the agent's interior best response at two periods in closed form, and unpacks the recovery–safety trade-off. Section 9 treats the organisation's joint design of strictness and duration and establishes the substitution between them. Section 10 shows that promoting a fixed quota of candidates is asymptotically equivalent to committing to a threshold. Section 15 discusses limitations and next steps. Proofs and numerical methodology are in the appendices.

Related Literature

Three literatures meet here. The first is the career-concerns tradition, which treats effort as the agent's instrument and infers ability from a sequence of output signals whose precision is given. We invert the focus: the agent picks how informative each signal will be. Cisternas (2018) extends the traditional effort model by introducing dynamic skill acquisition, showing the worker's incentive to invest in skill is weakened by the market's identification problem. Bonatti and Hörner (2017) shows that wage commitment mitigates the inefficient effort delay and sub-optimal effort under the assumption that effort levels are serially strategic substitutes. Hörner and Lambert (2021) constructs the optimal rating system on the planner's side and shows that higher effort might jeopardise later ratings. The agent side of our model arises from Hermalin (1993). Hermalin shows the agent's preference for sending pure-noise signals to hide the true ability under the normal-normal learning model. In our model, we let the organisation commit to a promotion threshold and probation length before the agent takes any actions. The committed promotion threshold adds another layer of strategic interaction; we inspect how the threshold affects the agent's behaviour and how to set the threshold optimally. The agent behaves differently across single- and multi-period games: in the single period, she hides when the threshold is lenient enough to make her initial position favourable, and reveals when the threshold demands it. In the multi-period setting, we find a tradeoff that involves two forces generated by asymmetric variances.

The second is up-or-out and tenure-track screening itself. The classical theoretical studies, Kahn and Huberman (1988) and Waldman (1990), ask why a firm would commit ex ante to dismiss workers who fail to clear a promotion bar instead of retaining them at a lower wage; their answers turn on firm-specific human capital and on signalling to competitors in the labour market. Ghosh and Waldman (2010) studies how the similarity between low- and high-level jobs and the demand for firm-specific capital drive the choice between standard promotion practices and up-or-out contracts. In our study, the organisation takes the up-or-out contract form as given and focuses purely on information extraction by optimising its contract strictness. We find that the range of feasible projects and average ability of agents drive the optimal threshold.

The third is contest theory with endogenous risk-taking. Section 10 replaces the deterministic threshold with a quota: promoting a fixed top-k of N cohort members. Hvide (2002) characterises risk choice when contestants can change the distribution parameters and shows excessive risk-taking in equilibrium. Genakos and Pagliero (2012) empirically document analogous risk-shifting in dynamic tournaments through interim-rank effects. They show awareness of an advantageous position hinders the agent's performance. The agent's best response in the extension coincides with the empirical evidence, from an information perspective, that the top half of the cohort generates only minimal information about their ability.

The dynamic intuition for the interior early-period optimal precision coincides with Section 6.2 of Meyer (1991): the opportunity to condition the later decision on the earlier signal drives the decision-maker away from the myopic rule. In our model, we further decompose this mechanism into two forces, each of which is conditional on one regime of the early-stage estimate.

2 Modelling

A risk-neutral organisation employs a risk-neutral agent whose latent ability η_i is fixed and drawn from a normal distribution at the beginning of the game. Neither party observes the ability directly, but they have a common prior belief that η_i is normally distributed with mean B and precision h_η:

$$\eta_i \sim \mathcal{N}(B,\ h_\eta^{-1}), \quad \text{equivalently} \quad \eta_i = B + \varepsilon_\eta,\ \ \varepsilon_\eta \sim \mathcal{N}(0, h_\eta^{-1}), \tag{1}$$

with $B \in \mathbb{R}$ and $h_\eta > 0$. It is equivalent to say the agent's ability is the prior mean of ability plus a noise term drawn from a zero-mean normal distribution with precision h_η.

Time in the probation period is discrete, $t = 1, 2, \ldots$. At the beginning of $t = 1$, the organisation hires the agent and commits to a promotion threshold η^* that cannot be revised. The agent receives tenure and a constant utility if and only if the posterior estimate of ability at the end of the last period weakly exceeds the threshold $\hat{\eta}_T \geq \eta^*$; otherwise, the agent is dismissed and gets zero utility. The length of probation is fixed at $T = 1$ throughout this section and is extended to multiple periods in Section 6.

In each probationary period the agent chooses a project p from a feasible set. A project is characterised by a pair (μ_p, h_p), both publicly observed: μ_p shifts mean output and h_p is the precision (inverse variance) of output noise. Conditional on project p,

$$x_p \ = \ \mu_p + \eta_i + \varepsilon_p, \qquad \varepsilon_p \sim \mathcal{N}(0,\ h_p^{-1}), \tag{2}$$

with ε_p independent across periods and across agents. The precision h_p is the agent's strategic instrument: a higher h_p makes the outcome $x_p - \mu_p = \eta_i + \varepsilon_p$ more informative in estimating the agent's true ability.

The project precision h_p captures the extent to which the project outcome reflects the agent's true ability. A high-precision project is one whose output is closely tied to the executor, such as a referee report, a course taught, or a software module with deterministic test coverage. A low-precision project is one where idiosyncratic noise dominates, such as a venture investment, a long-shot research direction, or a product launched into a thin market. In the latter case, inherent uncertainty implies that failures can be easily attributed to the environment, and posterior estimates cluster around the agent's prior ability. The agent's choice of project thus becomes an optimization of the variance of the posterior estimators.

Since the prior and the noise are normal, the posterior of η_i after t outcomes is normal as well. Based on DeGroot's Theorem 1 and standard results, the posterior estimator is the precision-weighted average

$$\hat{\eta}_{i,t} \ := \ \mathbb{E}_t[\eta_i] \ = \ \frac{h_\eta B \ + \ \sum_{s=1}^{t} h_{p(s)} \big(x_{i,s} - \mu_{p(s)}\big)}{h_\eta \ + \ \sum_{s=1}^{t} h_{p(s)}}, \tag{3}$$

and the variance of the posterior estimator equals $\big(h_\eta + \sum_{s=1}^{t} h_{p(s)}\big)^{-1}$. If an agent chooses a high-precision project, its outcome receives a large weight in computing the posterior estimator, while a low-precision project selection preserves more concentrated estimates around the agent's prior ability.

The organisation is risk-neutral and cares about the expected ability of tenured agents and the probability of tenure.

$$U(t) \ = \ \mathbb{E}_0\big[\, \hat{\eta}_{i,t} \mid \hat{\eta}_{i,t} \geq \eta^* \,\big] \ \Pr\!\big(\hat{\eta}_{i,t} \geq \eta^*\big) \ = \ \mathbb{E}_0\big[\, \hat{\eta}_{i,t}\, \mathbf{1}\{\hat{\eta}_{i,t} \geq \eta^*\}\,\big]\,, \tag{4}$$

In this section, we omit cost per period and hold the probation length fixed.

The rest of this section characterises the equilibrium outcomes. Section 3 derives the single-period best-response correspondences by solving the agent's project-choice problem at a fixed η^*; Section 4 returns to the organisation's problem and identifies the optimal threshold candidates for equilibrium characterisation; Section 5 characterises the conditions under which the informative equilibrium exists.

3 Agent's project choice

The agent receives a fixed prize upon promotion and zero otherwise; therefore, maximising expected utility is equivalent to maximising the probability of promotion. We solve for the agent's optimal project selection problem by taking the threshold η^* as given.

$$\max_{p \in \mathcal{P}} \ U_A(p) \ \equiv \ \Pr\!\big(\hat{\eta}_p \geq \eta^*\big). \tag{5}$$

Following equation (3), the posterior estimator at $t = 1$ conditional on a single project p is

$$\hat{\eta}_p \ = \ \delta_p\,(x_p - \mu_p) + (1 - \delta_p)\,B, \qquad \delta_p \ = \ \frac{h_p}{h_\eta + h_p}. \tag{6}$$

Because $x_p - \mu_p = \eta_i + \varepsilon_p = B + \varepsilon_\eta + \varepsilon_p$ is normal with mean B, $\hat{\eta}_p$ is normal with mean B and variance

$$\mathrm{Var}(\hat{\eta}_p) \ = \ \delta_p^2 \left(\frac{1}{h_\eta} + \frac{1}{h_p} \right) \ = \ \frac{h_p}{h_\eta(h_p + h_\eta)}. \tag{7}$$

Project choice therefore affects only the variance of posterior estimators, never their mean.

The probability of promotion follows with $\Phi(\cdot)$ denoting the standard normal cdf,

$$\Pr\!\big(\hat{\eta}_p \geq \eta^*\big) \ = \ 1 - \Phi\!\left(\frac{\eta^* - B}{\sqrt{\mathrm{Var}(\hat{\eta}_p)}} \right), \tag{8}$$

and the sign of $\eta^* - B$ pins down the agent's preferred dispersion. When $\eta^* < B$, the agent is ex ante favoured; the agent can keep the posterior tightly concentrated around the prior mean by choosing the project with the lowest precision. The high variance in the project outcome means it is weighted less in the posterior estimator; thus, even a terrible outcome would not pull the posterior below the threshold, because the organisation knows the terrible outcome is very likely to originate from the randomness of the project. When $\eta^* > B$, the agent is ex ante unfavoured and gains from widening the posterior distribution, so she chooses the highest-precision project to maximise the right-tail probability. Unlike in Hermalin (1993), where agents always want to hide information, in our model, the agent would like to reveal her ability as precisely as she can when she is in a disadvantageous position.

The boundary case $\eta^* = B$ requires care. For any $h_p > 0$, $\Pr(\hat{\eta}_p \geq B) = \frac{1}{2}$, so the agent is indifferent across all strictly positive precisions. If the feasible set includes the degenerate project $h_p = 0$, however, the posterior distribution collapses to a singleton at the prior mean ($\hat{\eta}_p \equiv B$) and the weak-inequality promotion rule promotes the agent with probability one. To ensure the existence of an equilibrium, we rule out this knife-edge by assuming projects always carry some information $\underline{h} > 0$. We also assume that the agent picks the project with the highest precision at $\eta^* = B$. The assumption plays the same role as the "accept when indifferent" selection in alternating-offer bargaining. Otherwise, the organisation could exploit highest-precision selection by increasing η^* marginally above B.

Lemma 3.1 gives the agent's best-response correspondence in a clean closed form, which allows us to solve the Stackelberg equilibrium by backward induction.

Lemma 3.1 (Agent's best-response correspondence)**.** *The agent chooses a project p from a finite set $\{(\mu_1, h_1), \ldots, (\mu_n, h_n)\}$ with $h_p \in [\underline{h}, \overline{h}]$ and μ_p publicly known. Under a tenure rule that grants tenure if and only if $\hat{\eta}_p \geq \eta^*$, the agent's best-response correspondence $h_p^*(\eta^*)$ satisfies*

1. *if $\eta^* < B$, $h_p^*(\eta^*) = \underline{h}$;*
2. *if $\eta^* > B$, $h_p^*(\eta^*) = \overline{h}$;*
3. *if $\eta^* = B$:*
   - (a) *for all $h_p > 0$, $\Pr(\hat{\eta}_p \geq B) = \frac{1}{2}$, so when $\underline{h} > 0$ the agent is indifferent over $[\underline{h}, \overline{h}]$;*
   - (b) *if $\underline{h} = 0$ and tenure is granted under the weak inequality, then $h_p = 0$ implies $\hat{\eta}_p \equiv B$ almost surely and $\Pr(\hat{\eta}_p \geq B) = 1$, so $h_p^*(B) = 0$ uniquely.*

For all $h_p > 0$, $\Pr(\hat{\eta}_p \geq \eta^*)$ is monotone in h_p with sign $\mathrm{sign}(\eta^* - B)$.

4 The organisation's preference over project precision

The organisation maximises its objective function (4) over the promotion threshold η^*.

We proceed in two steps. First, we solve a pure screening problem to identify the ex post optimal threshold by fixing the project precision h_p. Second, we solve the general problem where the agent chooses the project based on her best-response correspondence. We show that the organisation's problem collapses to a binary comparison between the ex post optimal threshold and the lowest threshold that motivates information revelation.

For fixed h_p, the posterior estimator satisfies $\hat{\eta}_p \sim \mathcal{N}(B, \sigma_b^2(h_p))$ where

$$\sigma_b(h_p) \;:=\; \sqrt{\frac{h_p}{h_\eta(h_p + h_\eta)}} \tag{9}$$

is the standard deviation of the posterior mean. Applying the truncated normal first-moment formula to (4),

$$U_p(h_p, \eta^*) \;=\; B(1 - \Phi(z)) + \sigma_b(h_p)\,\phi(z), \qquad z \;=\; \frac{\eta^* - B}{\sigma_b(h_p)}, \tag{10}$$

where Φ and ϕ denote the standard normal cdf and pdf. Differentiating in η^*,

$$\frac{\partial U_p}{\partial \eta^*} \;=\; -\,\frac{\eta^*}{\sigma_b(h_p)}\,\phi(z), \tag{11}$$

so U_p is single-peaked in η^* and maximised at $\eta^* = 0$. Setting $\eta^* = 0$ retains every agent whose posterior ability is positive, i.e. every agent expected to contribute positively to the organisation. This is the ex post optimal screening rule.

We now combine this ex post screening rule with the agent's response from Lemma 3.1, which selects high-precision projects iff $\eta^* \geq B$. Within the $\eta^* < B$ regime, the ex post optimum $\eta^* = 0$ lies inside the regime and is feasible. Within the $\eta^* \geq B$ regime, the ex post optimum is excluded and the constrained optimum is the corner $\eta^* = B$. The case $B \leq 0$ is trivial: at $\eta^* = 0$, the two cases are identical, with the agent choosing $\bar{h}$ at the ex post optimum, and no trade-off arises. We therefore focus on the non-trivial $B > 0$ case.

For the equilibrium analysis, we refer to the candidate ($\eta^* = B$, $h_p = \bar{h}$) as the *informative equilibrium* and refer to the candidate ($\eta^* = 0$, $h_p = \underline{h}$) as the *uninformative equilibrium*.

The organisation's problem becomes a binary choice between the uninformative and informative candidates:

$$\begin{aligned}
&\text{at uninformative threshold: } \eta^* = 0, \;\; h_p = \underline{h}\\
&\text{at informative threshold: } \eta^* = B, \;\; h_p = \bar{h}.
\end{aligned}$$

The organisation prefers high project precision, as the project outcome becomes more informative of the agent's ability than of luck. When estimating ability, the organisation puts a higher weight on the project outcome and a lower weight on the prior mean. This outcome-inclined estimator improves the organisation's utility by making the organisation more confident in the promoted agents' abilities, as the posterior estimator distribution generated by a high-precision project is closer to the true ability distribution.

Formally, we show that U_p is strictly increasing in h_p at $\eta^* = 0$ and at $\eta^* = B$. We have $\partial U_p/\partial h_p > 0$ at $\eta^* = 0$ and $\eta^* = B$ respectively:

$$\begin{aligned}
\frac{\partial U}{\partial h_p} &= \frac{\partial U}{\partial \sigma_b}\frac{\partial \sigma_b}{\partial h_p}\\
&= [\frac{\eta^* - B}{\sigma_b^2}(B\phi(z) - \sigma_b\phi'(z)) + \phi(z)]\frac{\partial \sigma_b}{\partial h_p} && (12)\\
\text{where } \frac{\partial \sigma_b}{\partial h_p} &= \sqrt{h_\eta}\,/\,[2\sqrt{h_p}\,(h_\eta + h_p)^{3/2}] > 0 && (13)\\
\frac{\partial U(\eta^* = 0)}{\partial h_p} &= \phi(z)\,\underbrace{\frac{\partial \sigma_b}{\partial h_p}}_{>0} > 0 && (14)\\
\frac{\partial U(\eta^* = B)}{\partial h_p} &= \underbrace{[\frac{\eta^* - B}{\sigma_b^2}(B\phi(z) + \sigma_b z\phi(z)) + \phi(z)]}_{>0 \text{ if } \eta^* \geq B \geq 0}\underbrace{\frac{\partial \sigma_b}{\partial h_p}}_{>0} > 0 && (15)
\end{aligned}$$

The uninformative candidate attains the ex post optimum at the cost of the least informative signal; the informative candidate allows a better estimator of ability at the cost of a Type-I error due to the high threshold. Which one is preferred depends on the feasible range of h_p, which we parametrise by a benchmark precision s and a dispersion parameter x.

5 The informative equilibrium

To better isolate the effects of project precision, we reframe the upper and the lower bounds of feasible project precision by a benchmark precision $s > 0$ and a dispersion $x \in [0, s)$, so that

$h_p \in [\underline{h}, \bar{h}]$ with $\underline{h} = s - x$ and $\bar{h} = s + x$. The standard deviation of the posterior mean $\sigma_b(\cdot)$ defined in (9) is strictly increasing in h_p, with $\sigma_b(0) = 0$ and $\lim_{h_p \to \infty} \sigma_b(h_p) = 1/\sqrt{h_\eta}$.

Substituting the two threshold candidates from §4 into the closed-form payoff (10) yields

$$U_p^\ell(x, s) \;:=\; U_p(\underline{h},\ 0) \;=\; B\,\Phi\Big(\tfrac{B}{\sigma_b(\underline{h})}\Big) + \sigma_b(\underline{h})\,\phi\Big(\tfrac{B}{\sigma_b(\underline{h})}\Big), \tag{16}$$

$$U_p^h(x, s) \;:=\; U_p(\bar{h},\ B) \;=\; \tfrac{B}{2} + \sigma_b(\bar{h})\,\phi(0). \tag{17}$$

Their difference,

$$F(x, s;\, B, h_\eta) \;:=\; U_p^h(x, s) - U_p^\ell(x, s), \tag{18}$$

measures the net benefit of the informative regime over the ex post optimum. The informative equilibrium is sustainable if and only if $F \geq 0$.

We inspect two extreme cases of $F(x, s;\, B, h_\eta)$ regarding x. When the precision range collapses to a singleton ($x \to 0$, $\underline{h} = \bar{h}$), the feasible project set becomes a singleton, and the informative candidate carries no information advantage over the uninformative candidate. Raising the threshold η^* from 0 to B only distorts screening; equivalently, $F(x^* = 0, s) < 0$ for any $B > 0$. At the opposite extreme, where the organisation extracts the maximum information advantage from setting the threshold at the prior mean ($x \to s$ with $s \to \infty$, so $\underline{h} \to 0$ and $\bar{h} \to \infty$), F approaches a strict upper bound that depends only on the prior mean B and ability precision h_η. Qualitatively, if the information advantage the organisation receives from a higher threshold outweighs the cost of suboptimal screening, the organisation is better off by motivating high-precision selection. Using the intermediate value theorem, we characterise the condition for the existence of an informative equilibrium in Proposition 5.1.

Proposition 5.1 (Existence of an informative equilibrium). *Let $h_\eta > 0$ and $B \in \left(0,\ \phi(0)\,\dfrac{2}{\sqrt{h_\eta}}\right)$. Then there exists a unique $s^*(B, h_\eta) \geq 0$ such that $\eta^* = B$ in the unique equilibrium if and only if $s > s^*(B, h_\eta)$ and $x \in (x^*(s; B, h_\eta),\, s)$, where $x^*(s; B, h_\eta)$ is the unique root of $F(x^*(s; B, h_\eta), s;\, B, h_\eta) = 0$.*

Remark 5.2 (Boundary case $B = 0$). When $B = 0$, the two equilibrium candidates are identical, so the equilibrium is unique.

Proof. **Step 1. Necessity of the bound on B.** The supremum of F over the feasible set is approached in the joint limit $x \to s$ and $s \to \infty$, equivalently, $\underline{h} \to 0^+$ and $\bar{h} \to \infty$, where the higher threshold extracts the largest information advantage. As $\underline{h} \to 0^+$, $\sigma_b(\underline{h}) \to 0$ and $B/\sigma_b(\underline{h}) \to +\infty$, so $\Phi\big(B/\sigma_b(\underline{h})\big) \to 1$ and $\sigma_b(\underline{h})\,\phi\big(B/\sigma_b(\underline{h})\big) \to 0$, giving $U_p^\ell \to B$. As $\bar{h} \to \infty$, $\sigma_b(\bar{h}) \to 1/\sqrt{h_\eta}$, giving $U_p^h \to B/2 + \phi(0)/\sqrt{h_\eta}$. Hence

$$\sup_{(s,\,x)} F \;=\; -\,\frac{B}{2} \;+\; \frac{\phi(0)}{\sqrt{h_\eta}}, \tag{19}$$

approached but not attained. For any informative equilibrium to exist the supremum must be strictly positive, so $B < 2\phi(0)/\sqrt{h_\eta}$.

Step 2. Existence of s^*. The map $s \mapsto \sigma_b(2s)$ is continuous and strictly increasing on $[0, \infty)$, with $\sigma_b(0) = 0$ and $\lim_{s \to \infty} \sigma_b(2s) = 1/\sqrt{h_\eta}$. Step 1 ensures $B/(2\phi(0)) < 1/\sqrt{h_\eta}$, so by the intermediate value theorem there is a unique $s^* \geq 0$ with $\sigma_b(2s^*) = B/(2\phi(0))$. The condition $s > s^*$ is therefore equivalent to

$$\sigma_b(2s) \;>\; \frac{B}{2\phi(0)}. \tag{20}$$

Step 3. For $s > s^*$, existence and uniqueness of $x^*(s)$.
We verify three properties of $F(\,\cdot\,, s)$ on $[0, s)$.

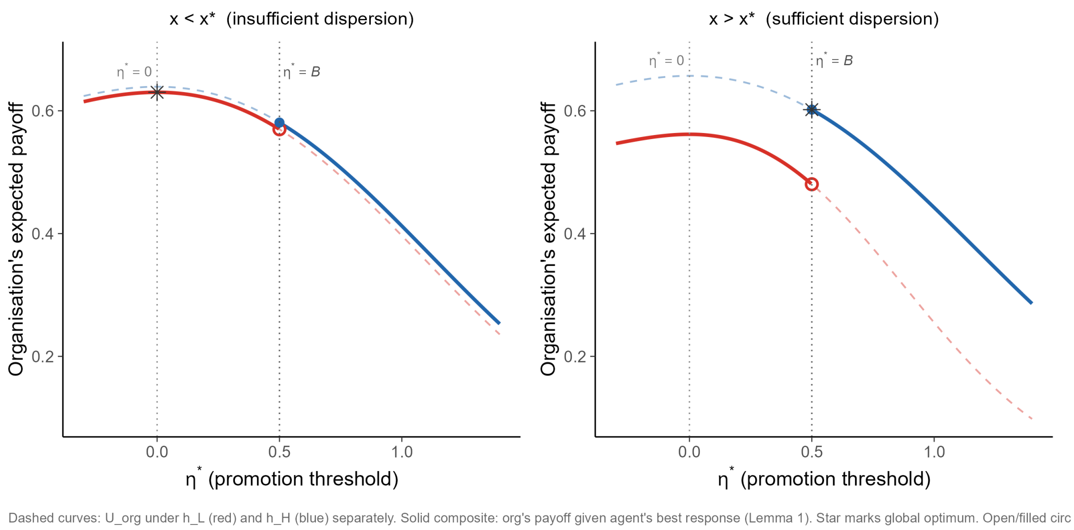

Figure 1: Organisation's payoff U_p versus the promotion threshold η^* given the agent's best response. The uninformative threshold $\eta^* = 0$ is the global maximum when $x < x^*$ (left, $x = 0.2$); the informative threshold $\eta^* = B$ is the global maximum when $x > x^*$ (right, $x = 1.5$). Parameters: $B = 0.5$, $h_\eta = 1$, $s = 2$.

(i) Continuity and monotonicity. F is continuous as a composition of continuous functions. By the argument in §4, U_p is strictly increasing in h_p at both $\eta^* = 0$ and $\eta^* = B$. Since $\bar{h} = s+x$ is increasing and $\underline{h} = s - x$ is decreasing in x, U_p^h is increasing and U_p^ℓ is decreasing in x, so F is strictly increasing in x.

(ii) $F(0, s) < 0$. At $x = 0$, $\underline{h} = \bar{h} = s$. Writing $t = B/\sigma_b(s) > 0$ and factoring,

$$F(0, s) = \sigma_b(s)\, g(t), \qquad g(t) := t\big[\tfrac{1}{2} - \Phi(t)\big] + \phi(0) - \phi(t). \tag{21}$$

Since $g(0) = 0$ and $g'(t) = \frac{1}{2} - \Phi(t) < 0$ for $t > 0$, we have $g(t) < 0$ on $(0, \infty)$, hence $F(0, s) < 0$.

(iii) $\lim_{x\to s^-} F(x, s) > 0$. Repeating the limit argument from Step 1 in $\underline{h}$ alone gives $U_p^\ell \to B$ as $\underline{h} \to 0^+$, hence

$$\lim_{x\to s^-} F(x, s) = -\frac{B}{2} + \sigma_b(2s)\,\phi(0). \tag{22}$$

For $s > s^*$, (20) makes this strictly positive.

By continuity, strict monotonicity in x, and opposite endpoint signs, the intermediate value theorem yields a unique $x^*(s) \in (0, s)$ with $F(x^*(s), s) = 0$. For $x \in (x^*(s), s)$, $F > 0$ and $\eta^* = B$ is uniquely optimal; for $x < x^*(s)$, $F < 0$ and $\eta^* = 0$ is uniquely optimal; at $x = x^*(s)$ the organisation is indifferent, and the open interval excludes this knife-edge.

Step 4. For $s \le s^*$, no informative equilibrium. When $s \le s^*$, the monotonicity of $\sigma_b(2\cdot)$ from Step 2 and the defining equation of s^* give $\sigma_b(2s) \le B/(2\phi(0))$, hence $\lim_{x\to s^-} F(x, s) \le 0$. Combined with $F(0, s) < 0$ and strict monotonicity in x, $F(x, s) < 0$ for all $x \in [0, s)$. □

Figure 1 shows that the organisation prefers the informative equilibrium when the precision dispersion is large enough ($x > x^*$), and prefers the uninformative equilibrium when x is small. The reason is that by increasing x, we are reducing $\underline{h}$ and increasing $\bar{h}$, which makes the informative candidate more favourable and weakens the uninformative candidate.

Figure 2 illustrates the parametric regime for the existence of an informative equilibrium, as characterised in Proposition 5.1. The shaded area corresponds to the parameters that sustain the informative equilibrium. The informative equilibrium only holds when s and x exceed their critical values.

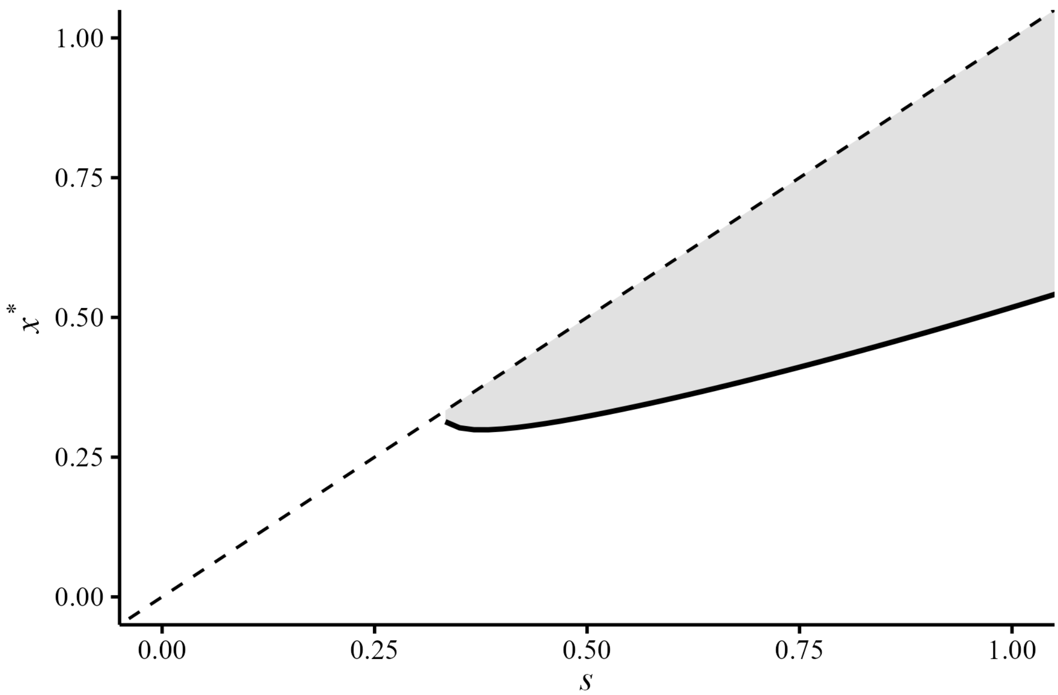

Figure 2: Parametric regime in which the informative equilibrium exists under $B = 0.5$ and $h_\eta = 1$.

Effect of the prior mean. Differentiating $F(x^*(s), s; B, h_\eta) = 0$ implicitly gives, for $B > 0$,

$$\frac{\partial x^*}{\partial B} = -\frac{\partial F/\partial B}{\partial F/\partial x} > 0, \tag{23}$$

since $\partial F/\partial B = \frac{1}{2} - \Phi(B/\sigma_b(\underline{h})) < 0$ and $\partial F/\partial x > 0$ from Step 3(i). A higher prior mean of ability does two things at once: it widens the gap between the ex post optimal threshold and the informative threshold, raising the cost of distortion, and it shrinks the proportion of agents the screen actually filters, weakening the value of information. Both forces require the greater ability revelation in the informative equilibrium to compensate for the larger Type-I error.

Effect of prior precision. A lower prior precision h_η raises the upper bound on B and lowers s^*, since the right-hand side $B/(2\phi(0))$ falls relative to $1/\sqrt{h_\eta}$. The agent's ability is less concentrated around the prior mean, so the organisation suffers less from setting a high threshold, which makes the informative equilibrium easier to sustain.

Effect of benchmark precision. The effect of s on x^* is non-monotonic. Near the existence boundary $x^* \to s^{*-}$ the agent could hide their ability very easily as $\underline{h} \to 0$. The posterior estimator distribution for the uninformative candidate is concentrated around the prior mean, so a marginal increase in s has a limited effect on screening: the **increase in** $\Pr(\hat{\eta} < 0)$ is smaller than the gain for its informative counterpart. Appendix E gives the global characterisation and a numerical example.

6 Multi-Period Probation

7 Overview

In the single period model, the informative equilibrium tells us that the organisation can motivate better ability revealing by setting a higher threshold. In practice, the up-or-out contracts vary not only in how selective the promotion threshold is, but also in how long they expect the agents to produce impressive results. In this section, we extend the model by allowing the organisation to incur a constant cost per period to commit to a longer probation period, so that the agent could repetitively choose a project and observe the outcome in each period of the probation. This section studies how the organisation jointly designs the promotion threshold and the probation length, and whether longer probation complements stricter promotion.

In Section 2, the organisation's trade-off was clear, and the agent's best response was simple and binary: $\eta^* < B$ induced the low-precision project $\underline{h}$, while $\eta^* \geq B$ encouraged the high-precision project selection $\bar{h}$. The organisation's optimization problem reduced to a comparison of the ex-post optimum and the minimum threshold required to motivate ability revelation. Multi-period probation complicates the problem by adding two forces to the agent's problem and making the organisation's problem continuous. With $T \geq 2$, the agent can adapt her period-t project to the signals she has already observed; this adaptation makes the optimal early project selection interior, provided $[\underline{h}, \bar{h}]$ is sufficiently wide.[1] The agent's period-1 best response can be interior, falling strictly between $\underline{h}$ and $\bar{h}$ for a range of thresholds including $\eta^* = B$.

In the multi-period setup, the normality of the posterior estimator after each outcome realisation carries through, and the organisation's problem becomes continuous in both the threshold η^* and the probation length T. A higher threshold raises the probability of motivating the agent to select a higher-precision project, but at the cost of failing to promote more agents with positive ability. Since the agent's best response admits no closed form for general $T > 1$, we bound the organisation's payoff in each regime by a constant-precision benchmark — $\bar{h}$ in the informative regime, $\underline{h}$ in the uninformative regime — and characterise the duration–strictness substitution analytically under those bounds. By duration–strictness substitution we mean the trade-off between probation length T and threshold strictness η^* in the organisation's joint design: how much longer probation is needed to compensate for a more lenient threshold.

The remainder of this section solves the agent's side of the joint design, working through setup, terminal best response, closed-form analysis at $T = 2$, the two-forces decomposition, and a qualitative extension to general T. Three results anchor the analysis: a clean terminal best response (Lemma 8.1); a closed-form period-1 objective at $T = 2$, $\eta^* = B$, established to be strictly quasi-concave with a unique interior maximiser (Lemmas 8.2–8.3); and qualitative observations on the agent's best response at general T. These results feed into the organisation's joint design of (η^*, T) in Section 9.

8 The Agent under Multi-Period Probation

8.1 Setup

The agent is hired in period $t = 1$ and faces a probation of length $T \in \mathbb{N}$. Latent ability $\eta_i \sim \mathcal{N}(B, 1/h_\eta)$ is unknown to both parties. In each period $t = 1, \ldots, T$ the agent chooses a project precision $h_t \in [\underline{h}, \bar{h}]$ *before* observing the period's outcome. The period-t outcome

$$x_t \; = \; \eta_i + \mu_t + \varepsilon_t, \qquad \varepsilon_t \mid h_t \sim \mathcal{N}(0,\, 1/h_t), \tag{24}$$

realises at the end of the period and is publicly observed. Bayesian updating yields a posterior mean $\hat{\eta}_t$ with cumulative posterior precision

$$H_t \; = \; h_\eta + \sum_{s=1}^{t} h_s, \qquad H_0 := h_\eta, \tag{25}$$

and posterior variance $1/H_t$. At the end of period T the organisation promotes the agent if and only if $\hat{\eta}_T \geq \eta^*$. Similar to the derivation in the single-period model, we can write the posterior estimator of the next period conditional on the current period estimate as

$$\hat{\eta}_{t+1} \mid \hat{\eta}_t \sim \mathcal{N}(\hat{\eta}_t,\, \frac{h_{t+1}}{H_t(H_t + h_{t+1})}). \tag{26}$$

[1]This finding coincides with Meyer (1991): a learner who can choose how to weight later observations on the basis of earlier ones generally departs from myopic decision rules.

The agent maximises the promotion probability conditional on the current estimate of her ability and the record of previous projects. We denote the history by $\mathcal{F}_{t-1} := (h_1, x_1, \ldots, h_{t-1}, x_{t-1})$. The agent's current stage objective becomes

$$\mathcal{P}_t \;:=\; \Pr(\hat{\eta}_T \geq \eta^* \,|\, h_t,\; \mathcal{F}_{t-1};\,), \tag{27}$$

In particular, with $T = 2$, the agent's objective after observing the first outcome is

$$\mathcal{P}_2 \;=\; \Pr(\hat{\eta}_2 \geq \eta^* \,|\, h_2; h_1, x_1). \tag{28}$$

8.2 Last-Period Best Response

The dynamic problem has a clean terminal anchor.

Lemma 8.1. *In period T, the agent's best response is the one-period rule of Lemma 3.1 applied to the period-$(T-1)$ posterior $(\hat{\eta}_{T-1}, H_{T-1})$ in place of the prior (B, h_η). Specifically, the agent chooses $\bar{h}$ if $\hat{\eta}_{T-1} \leq \eta^*$, $\underline{h}$ if $\hat{\eta}_{T-1} > \eta^*$, and $\bar{h}$ at the tie $\hat{\eta}_{T-1} = \eta^*$.*

Proof. At $t = T$ the continuation value is degenerate: only the period-T signal can move the posterior across the threshold. Conditional on $\hat{\eta}_{T-1}$, $\hat{\eta}_T$ is normal with mean $\hat{\eta}_{T-1}$ and variance $h_T/[H_{T-1}(H_{T-1} + h_T)]$. The argument of Lemma 3.1 then applies verbatim with $(\hat{\eta}_{T-1}, H_{T-1})$ replacing (B, h_η). □

Lemma 8.1 pins down the terminal node of the dynamic programme. We decompose the agent's early-period decision-making into two forces: a *recovery force* and a *safety force* that the agent relies on to maximise the promotion probability.

8.3 Two-Period Closed Form at $\eta^* = B$

We work out the case $T = 2$, $\eta^* = B$ in closed form; the results provide intuition for the general intractable case. At the start of period 1 the agent selects h_1 to maximise the probability that, after two signals, $\hat{\eta}_2 \geq \eta^*$, anticipating the optimal period-2 decision rule under Lemma 8.1.

The ex-ante promotion probability decomposes around the threshold the agent's period-1 posterior must cross to switch her period-2 project:

$$\Pr(\hat{\eta}_2 \geq \eta^*) = \Pr(\hat{\eta}_2 \geq \eta^* \mid \hat{\eta}_1 < \eta^*) \Pr(\hat{\eta}_1 < \eta^*) + \Pr(\hat{\eta}_2 \geq \eta^* \mid \hat{\eta}_1 > \eta^*) \Pr(\hat{\eta}_1 > \eta^*) \tag{29}$$

$$= \int_{-\infty}^{\eta^*} \Pr(\hat{\eta}_2 \geq \eta^* \mid \hat{\eta}_1)\, f_1(\hat{\eta}_1)\, d\hat{\eta}_1 + \int_{\eta^*}^{\infty} \Pr(\hat{\eta}_2 \geq \eta^* \mid \hat{\eta}_1)\, f_1(\hat{\eta}_1)\, d\hat{\eta}_1, \tag{30}$$

where f_1 is the density of $\hat{\eta}_1 \sim \mathcal{N}(B, \sigma_1^2)$ with $\sigma_1^2 = h_1/[h_\eta(h_\eta + h_1)]$.

If the first period estimate lies below the threshold, the agent will play $\bar{h}$ in period 2, so the conditional standard deviation of the last period posterior estimator distribution is $\sigma_2^H = \sqrt{\bar{h}/[H_1(H_1 + \bar{h})]}$. If the first period estimate lies above the threshold, she plays $\underline{h}$, giving $\sigma_2^L = \sqrt{\underline{h}/[H_1(H_1 + \underline{h})]}$. Standardising $z = (\hat{\eta}_1 - B)/\sigma_1$ and denoting $r^H := \sigma_1/\sigma_2^H$, $r^L := \sigma_1/\sigma_2^L$, equation (30) becomes

$$\mathcal{P}(h_1; \eta^*) \;=\; \int_{-\infty}^{z^*} \Phi(r^H(z - z^*))\varphi(z)\, dz \;+\; \int_{z^*}^{\infty} \Phi(r^L(z - z^*))\varphi(z)\, dz, \tag{31}$$

with $z^* = (\eta^* - B)/\sigma_1$.

At $\eta^* = B$ we have $z^* = 0$, the integrand on the negative half-line folds onto the positive one by $\varphi(-u) = \varphi(u)$ and $\Phi(-x) = 1 - \Phi(x)$, and Sheppard's orthant formula for bivariate normals (Sheppard 1898) delivers a closed form.

Table 1: Optimal first-period precision h_1^* as a function of the promotion threshold η^*, for two values of $\bar{h}$. Other parameters: $\underline{h} = 0.5$, $B = 1$, $h_\eta = 1$.

η^*	0.90	0.95	1.00	1.05	1.10
$h_1^*(\bar{h} = 2)$	0.5000	0.5000	0.5854	0.7256	0.8973
$h_1^*(\bar{h} = 5)$	0.5508	0.6582	0.7710	0.8905	1.0184

Lemma 8.2 (Period-1 promotion probability at $\eta^* = B$)**.** *Under $T = 2$ and $\eta^* = B$, the agent's period-1 objective admits the closed form*

$$\mathcal{P}(h_1; B) \;=\; \frac{1}{2} \;+\; \frac{1}{2\pi}\Big[\arctan\!\big(r^L(h_1)\big) - \arctan\!\big(r^H(h_1)\big)\Big], \tag{32}$$

where

$$\big(r^L(h_1)\big)^2 \;=\; \frac{h_1\,(h_\eta + h_1 + \underline{h})}{h_\eta\,\underline{h}}, \qquad \big(r^H(h_1)\big)^2 \;=\; \frac{h_1\,(h_\eta + h_1 + \bar{h})}{h_\eta\,\bar{h}}. \tag{33}$$

Lemma 8.3. *The function $\mathcal{P}(h_1; B)$ defined in* (32)–(33) *is strictly quasi-concave on $h_1 \in (0, \infty)$ and admits a unique maximiser $h_1^* \in (0, \infty)$.*

The full derivation, which uses Sheppard's orthant formula for bivariate normals together with trigonometric identities, is presented in Appendix B.

The closed form allows for numerical evaluation. Take $B = 1$, $h_\eta = 1$, $\underline{h} = 0.5$, $\bar{h} = 2$ as a baseline, and report the optimal first-period precision h_1^* across five values of η^* and two values of $\bar{h}$:

Table 1[2] reports the optimal first-period precision h_1^* for η^* under two values of $\bar{h}$ with multiple thresholds η^*. In the first row, which features a smaller precision range $[\underline{h}, \bar{h}] = [0.5, 2]$, the myopic rule still applies when $\eta^* < B$. The optimal precision for higher thresholds implies that, in a two-period setting, it is more costly in terms of screening for an organisation to motivate a high-precision project during the early stages of probation.

The second row, with a wider precision range $[\underline{h}, \bar{h}] = [1, 5]$, shows that the optimal first-period precision is strictly (and non-trivially) above $\underline{h}$ for $\eta^* < B$, as agents are willing to select higher-precision projects. The trade-off between the recovery and safety forces explains this variation.

8.4 The Two Forces

In equation (30), $\mathcal{P}(h_1; B)$ is partitioned into two integrals: an event in which the period-1 estimate lands below the threshold, and another event in which it exceeds the threshold. In these two events, the agent selects the projects with the highest precision and the lowest precision, respectively. In the former case, the high-precision project generates a thicker right tail, which *recovers* her chance of promotion; we write the recovery probability as $\Pr(\hat{\eta}_2 \geq \eta^* \mid \hat{\eta}_1 < \eta^*)$. In the latter case, the low-precision project generates a final-estimator distribution concentrated around the already-favourable $\hat{\eta}_1$, and its thin left tail *safeguards* the promotion probability by minimizing the dismissal probability $\Pr(\hat{\eta}_2 < \eta^*)$; we write the corresponding survival probability as $\Pr(\hat{\eta}_2 \geq \eta^* \mid \hat{\eta}_1 > \eta^*)$.

In the rest of the section, we refer to $\Pr(\hat{\eta}_2 \geq \eta^* \mid \hat{\eta}_1 < \eta^*; h_2 = \bar{h})$ as the *recovery force* and $\Pr(\hat{\eta}_2 \geq \eta^* \mid \hat{\eta}_1 > \eta^*; h_2 = \underline{h})$ as the *safety force*. Graphically, the recovery force is the area under the solid distribution to the right of the threshold line in Figure 3, while the safety force is the area under the dashed distribution to the right of the threshold line. These two forces respond differently to h_1, and their trade-off shapes the agent's optimal first-period precision.

[2] Replication code for the numerical results in this paper is available at the author's website, `https://diwen-s.github.io/`.

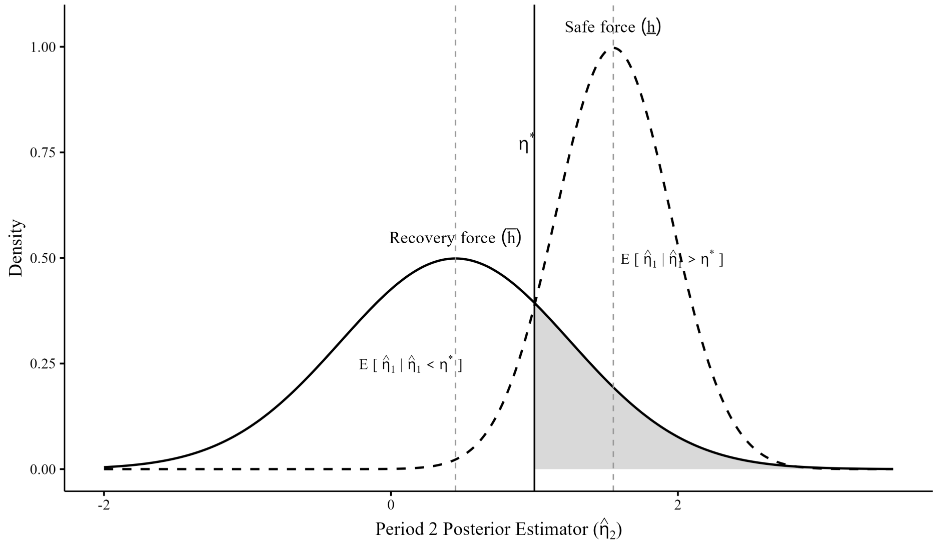

Figure 3: Density of the period-2 posterior estimator $\hat{\eta}_2$, evaluated at the expected period-1 estimator within each region: $E[\hat{\eta}_1 \mid \hat{\eta}_1 < \eta^*]$ for the solid (recovery, $h_2 = \bar{h}$) curve and $E[\hat{\eta}_1 \mid \hat{\eta}_1 > \eta^*]$ for the dashed (safety, $h_2 = \underline{h}$) curve, for a given h_1. The solid vertical line marks the promotion threshold; the dashed verticals mark these expected period-1 estimators.

The first-period precision enters both the mean and variance of the final estimator's distribution. A higher h_1 increases the variances of both distributions and pushes their means further from the prior mean; such effects weaken the recovery force and strengthen the safety force.

Recall the final-period estimator variance in (26): the cumulative precision from completed projects, H_1, enters the denominator, so a higher first-period precision makes later distributions more concentrated. Concentrated distributions reduce the recovery force, as it becomes less likely to generate the exceptional outcomes needed to recover promotion in the final stage, while they strengthen the safety force.

Meanwhile, the first-period precision also enters the variance of the first-period estimator $\hat{\eta}_1$. Following the standard result that the variance of the estimator increases with the precision of the current-stage project, a higher h_1 disperses the distribution of the first-stage estimator, which implies more dispersed means for the later-stage estimator conditional on passing or failing the threshold. Agents must produce even more exceptional outcomes to recover promotion if they fail to reach the threshold in their early probation stage, while the same force locks in the advantage for agents who pass the threshold.

Both the mean and variance effects of h_1 increase the safety force and reduce the recovery force. Equivalently, the marginal effect of h_1 consists of a marginal cost from reducing the recovery force and a marginal gain from a stronger safety force. At low levels of h_1, the marginal gain to the safety force exceeds the marginal cost of weakening the recovery force, such that $\mathcal{P}'(h_1) > 0$; as h_1 grows, the recovery loss accelerates and eventually dominates, leading to $\mathcal{P}'(h_1) < 0$. The optimal h_1^* balances these two effects; graphically, h_1^* is the h_1 at which the two densities in Figure 3 intersect at η^*, where marginal cost equals marginal gain.

Using the MR = MC toolkit, Table 1 shows how the upper bound precision $\bar{h}$ enters through the recovery forces: a larger $\bar{h}$ increases the variance for below-threshold agents in the final period. The recovery force is stronger and weakens more slowly as h_1 rises. A lower marginal cost allows the agent to "spend" more precision in the earlier period to achieve a stronger safety force. The second row of Table 1 confirms that this intuition applies not only for $\eta^* \geq B$ but also for $\eta^* < B$. While the two forces are intuitive, it remains difficult to derive parameterised equilibrium conditions without a closed-form expression of the best-response correspondence.

Table 2: Optimal first-period precision at $\eta^* = B$, single-parameter perturbations (continuous solver).

Perturbation	h_η	$(\underline{h}, \bar{h})$	h_1^*	Interior
baseline	1.0	(0.50, 2.00)	0.5854	yes
small h_η	0.5	(0.50, 2.00)	0.5303	yes
large h_η	2.0	(0.50, 2.00)	0.6594	yes
low $\underline{h}$	1.0	(0.25, 2.00)	0.4133	yes
high $\bar{h}$	1.0	(0.50, 5.00)	0.7710	yes

We now turn to comparative statics of the interior maximiser h_1^* itself. While there is no simple closed-form solution for optimal precision h_1, our two-force decomposition and the equivalent MR = MC framework in Figure 3 allow us to understand comparative statics. Numerical results in Table 2 are simulated at the baseline calibration $B = 1$, $h_\eta = 1$, $\underline{h} = 0.5$, $\bar{h} = 2$, and we examine how ability precision and the range of project precision affect the optimal first-period precision.

Remark 8.4 (B-invariance at $\eta^* = B$). Neither r^L nor r^H in (33) depends on the prior mean B. By Lemma 8.2, the agent's objective $\mathcal{P}(h_1; B)$ at $\eta^* = B$ is a function of $(h_\eta, \underline{h}, \bar{h})$ alone, and so is its maximiser h_1^*.

Effect of ability precision. The prior precision enters the standard deviations of both the period-1 estimator σ_1 and the period-2 conditional estimator σ_2^j, so a higher h_η reshapes the safety and recovery forces simultaneously. The net reshaping pushes MR above MC at the previous h_1^*. The agent absorbs the wedge by raising first-period precision until MR = MC is restored at a higher value, tracing $h_1^* = 0.530 \to 0.586 \to 0.659$ as h_η doubles twice.

Effect of the project spread. The low $\underline{h}$ row shows how the agent reacts to a lower minimum precision, and the intuition fits well with the MR = MC framework. In contrast to the effect of $\bar{h}$ on the recovery force, a lower $\underline{h}$ enhances the safety force. Because a lower $\underline{h}$ makes the distribution of the final estimator given successful early projects more concentrated, the marginal revenue shrinks at the old h_1^*. Thus, the agent reduces the first-period precision, because the stronger safety force now better locks in early success. The last row confirms Table 1 and also shows that at $\eta^* = B$ the optimal first-period precision is B-invariant.

8.5 General T: A Qualitative Argument

A qualitative argument for the general multi-period environment arises from the $T = 2$ analysis. Lemma 8.3 establishes that the agent's optimal first-period allocation is interior: she neither reveals her ability with the highest precision early ($h_1 = \bar{h}$) nor hides entirely ($h_1 = \underline{h}$). The simplification that leads to the two-period closed form does not extend beyond $T = 2$. The agent's dynamic programme at $T \geq 3$ does not admit a tractable closed-form solution, and a quantitative characterisation of h_t^* at general T is left to future work. We gather two qualitative observations that frame the role of the agent's general-T strategy in the organisation's joint-design problem of Section 9.

The first observation concerns the variance asymmetry that drives the recovery–safety trade-off. An agent prefers a thick right tail to recover when she falls below the threshold and a concentrated distribution to lock in success if she has an impressive early stage outcome. The forces that deliver the interior optimum at $T = 2$ are structurally present at every $T \geq 2$. By the same logic, the boundary agent with true ability $\eta_i = B$ earns a promotion probability strictly above one half under optimal play at every finite T, since she strictly prefers her interior allocation to either corner whenever $[\underline{h}, \bar{h}]$ is wide enough.

Another observation comes from a dynamic programming perspective based on the interior solution. The general multi-period optimal precision could be viewed as dynamic programming over the "consumption of precision" in each period. Based on the interior solution in the $T = 2$ case, and the asymmetric variance argument in the first observation, we conjecture that the agent's optimal precision is interior in the multi-period setting and closer to the lower bound. Given the highly qualitative nature of the conjecture, in the analysis of the organisation in Section 9, we only use a weaker version of the observation: the agent's cumulative optimal precision across periods is weakly less than the cumulative precision that arises when the agent commits to the highest precision in every period, which holds automatically.

9 The Organisation's Joint Design

The interior early-period best response characterised in Section 8 admits no closed form, so the organisation's joint problem in (η^*, T) does not either. The decision variables are nonetheless continuous: the binary project selection in the final period and the normality of the posterior estimator carry through to general T. This lets us bound each regime's payoff by a constant-precision benchmark and characterise the joint design analytically under those bounds.

While the analytical solution is unavailable, we can still conduct a regime comparison between the most informative project selection and the least informative one. The equilibrium in the multi-period setting, if it exists, must lie between these bounds. To restore tractability, we assume T is continuous, which could be interpreted that the agent selects a project and sees the outcome for $[T-1]$ periods, and the last period's outcome has a higher weight $1 + T - [T]$ in the posterior updating. This interpretation allows us to use standard concavity tools without violating the best-response in the last period.

In the uninformative regime $\eta^* = 0$ with $B > 0$, we assume that the agent plays $\underline{h}$ in every period: it is the extended myopic rule from the one-period best response in Lemma 3.1. In the informative regime $\eta^* = B$, we assume the candidate plays $\bar{h}$ in every period, the strategy that yields the largest cumulated precision. Two extreme cases serve as the upper and lower bounds of project precision accumulation.

We establish three results: a sufficient condition (SC) on $(B, h_\eta, \underline{h})$ under which the uninformative gross payoff is concave in T and admits a unique finite T^*; an asymptotic regime gap $F_\infty \leq 0$ that rules out indefinite probation; and a duration–strictness substitution that pins the optimal probation on one side of a critical length T^{**} determined by the per-period cost c.

These bounds preserve the qualitative substitution between η^* and T^*. High cost shortens probation and pushes the organisation toward strict thresholds; low cost extends probation and favours lenient ones.

9.1 Regime Payoffs

Under the committed and constant precision h across T periods, posterior precision accumulates to $h_\eta + Th$, and posterior mean at the last period is normally distributed with variance

$$\sigma_T^2(h) := \frac{Th}{h_\eta(h_\eta + Th)}. \tag{34}$$

Applying the truncated-normal expectation formula at the respective thresholds, the organisation's utility in the two regimes is

$$U_I(T) := \frac{B}{2} + \phi(0)\,\sigma_T(\bar{h}) - c(T-1), \tag{35}$$

$$U_U(T) := B\,\Phi\Big(\frac{B}{\sigma_T(\underline{h})}\Big) + \sigma_T(\underline{h})\,\phi\Big(\frac{B}{\sigma_T(\underline{h})}\Big) - c(T-1). \tag{36}$$

Equations 35 and 36 are multi-period analogues of the single-period expressions 17 and 16, with an extra cost term $c(T-1)$. Given our focus on information instead of effort and wages, the per-period cost c captures the cost of further screening after the first period. Thus, if the cost of a longer period is too high for the organisation, there is an outside option of setting

$T = 1$, which is exactly the single-period case. Since $c(T-1)$ enters both expressions identically, the *regime gap*

$$F_T := U_I(T) - U_U(T) \tag{37}$$

is independent of c: it determines which regime the organisation prefers at a given probation length, while c determines the optimal length within each regime.

9.2 Optimal Probation Length

Each regime's gross payoff is strictly concave in T under a mild restriction on parameters, ensuring a unique finite optimum.

Proposition 9.1 (Unique finite optimum)**.** *Let $c > 0$.*

1. Informative regime. *$U_I(T)$ is strictly concave in T and admits a unique finite optimal probation length T_I^*.*
2. Uninformative regime. *If*

$$B^2 h_\eta \; < \; \frac{(\underline{h}/h_\eta)\,(1 + 4\,\underline{h}/h_\eta)}{1 + \underline{h}/h_\eta}, \tag{SC}$$

then $U_U(T)$ is strictly concave in T and admits a unique finite optimal probation length T_U^*.

Condition (SC) ensures that the least informative project is sufficiently distinct from pure noise. If the uninformative outcome is similar to the noise, in that tiny regime, marginally extending the probation makes a limited contribution to screening. This is the same force that drives the non-monotonic effect of s on x^* in the one-period model. Full derivation in Appendix B.

The first-order conditions characterise the interior optimum. From $\sigma_T'(h) = h\sqrt{h_\eta}\,/\,[2\sqrt{Th}\,(h_\eta + Th)^{3/2}]$, the optimal lengths T_I^* and T_U^* solve

$$\phi(0)\,\sigma'_{T_I^*}(\bar{h}) = c, \tag{38}$$

$$\phi\Big(\frac{B}{\sigma_{T_U^*}(\underline{h})}\Big)\,\sigma'_{T_U^*}(\underline{h}) = c. \tag{39}$$

Each left-hand side is strictly positive, continuous, and strictly decreasing in T, so each condition pins down a unique solution.

Proposition 9.2 (Failure-regime cost thresholds)**.** *Multi-period probation arises in each regime exactly when the per-period cost is sufficiently small. Defining*

$$c_T^I := \phi(0)\,\sigma_2'(\bar{h}), \qquad c_T^U := \phi\Big(\frac{B}{\sigma_2(\underline{h})}\Big)\,\sigma_2'(\underline{h}), \tag{40}$$

we have $T_I^* \geq 2 \iff c \leq c_T^I$ and $T_U^* \geq 2 \iff c \leq c_T^U$.

Below these thresholds the marginal informational gain at $T = 2$ exceeds the per-period cost, so the organisation extends probation beyond a single period; above them the organisation prefers the single-period outside option. Combined with Proposition 9.4, (c_T^I, c_T^U) characterise the cost ranges in which each regime supports a multi-period equilibrium. Proof in Appendix B.

To pin down the organisation's preference at a given probation length, we analyse the extreme cases of probation and argue that at a small T the organisation prefers the informative candidate, while at a large T it prefers the uninformative one. One extreme case is $T \to \infty$, where the cumulated precision becomes infinite and the distribution of the posterior estimator converges to the true distribution of ability: the organisation observes the agent's true ability directly. In this limit, a high threshold does not generate any information advantage, so the uninformative candidate with less screening distortion wins. Formally, $F_T \to F_\infty < 0$, follows from the same derivation as Step 3(ii) of Proposition 5.1.

If the conditions of Prop 5.1 hold so that the informative candidate is strictly preferred at $T = 1$, and the per-period cost lies in the range $c < \min c_T^I, c_T^U$ that supports multi-period probation in both regimes, we obtain a critical probation length: the informative regime dominates for shorter probations and the uninformative regime dominates for longer ones. We confirm this intuition by establishing that the regime gap is strictly quasi-concave in T, so the transition occurs exactly once.

9.3 Duration–Strictness Substitution

Lemma 9.3 (Concavity of the regime gap)**.** *Let $\sigma_L := \sigma_T(\underline{h})$ and define $\tilde{F}(\sigma_L) := F_T$, treating $\sigma_H = \sigma_T(\bar{h})$ as a function of σ_L via the common dependence on T. Then $\tilde{F}$ is strictly concave in σ_L on $(0,\, 1/\sqrt{h_\eta})$, and consequently F_T is strictly quasi-concave in T.*

The proof expresses $\sigma_H = \psi(\sigma_L)$ in closed form and shows that the derivative $\tilde{F}'(\sigma_L) = \phi(0)\psi'(\sigma_L) - \phi(B/\sigma_L)$ is strictly decreasing: $\phi(0)\psi'(\sigma_L)$ decreases because ψ' is strictly decreasing in σ_L, while $\phi(B/\sigma_L)$ is strictly increasing. Full proof in Appendix B.

Proposition 9.4 (Duration–strictness substitution)**.** *Let $B \in (0,\, 2\phi(0)/\sqrt{h_\eta})$ and suppose the precision dispersion x is large enough that $F_1 > 0$. Then there exists a unique $T^{**} > 1$ such that $F_{T^{**}} = 0$, $F_T > 0$ for all $T \in [1,\, T^{**})$, and $F_T < 0$ for all $T > T^{**}$.*

Proof. By Lemma 9.3, $F_T = \tilde{F}(\sigma_L(T))$ is quasi-concave in T, so the set $A := \{T \geq 1 : F_T \geq 0\}$ is an interval. Since $F_1 > 0$, A contains 1 and is nonempty; since $F_T \to F_\infty < 0$, A is bounded above. Setting $T^{**} := \sup A$ and invoking continuity gives $F_{T^{**}} = 0$. Strict concavity of $\tilde{F}$ with $F_1 > 0 = F_{T^{**}}$ implies $F_T > 0$ on $(1,\, T^{**})$, confirming uniqueness. □

Duration and strictness are *substitutes* in contract design. A longer duration allows precision accumulation, resulting in a better estimator as time goes on, but incurs the probation extension cost $c(T-1)$. A higher threshold motivates high precision at the expense of sub-optimal screening. At short horizons the gap in posterior precision between $\bar{h}$ and $\underline{h}$ projects is large, and the informational gain from inducing $\bar{h}$ outweighs the screening cost of $\eta^* = B$. As the horizon lengthens, the information advantage shrinks, since the cumulated precision enters the denominator of the variance. When the probation length is longer than T^{**}, the information advantage does not compensate for the loss from sub-optimal screening, and the uninformative regime dominates. The crossing point T^{**} depends only on $(B, h_\eta, \underline{h}, \bar{h})$; the per-period cost c then selects which regime the organisation operates in by determining the optimal horizon within each. High per-period cost c makes long probation less attractive, so the optimal probation T^* lies below T^{**}, favouring strict thresholds and short evaluation. Cheap extension pushes T^* above T^{**}, favouring lenient thresholds and extended screening. This provides a structural account of cross-institutional variation. Organisations facing high costs of maintaining probationary length, such as the cost of recurring senior review or the opportunity cost of deferring permanent assignment, optimally adopt strict, short-horizon promotion rules; organisations with low probation costs can afford extended evaluation under lenient standards.

10 Extension: competition for a fixed number of slots

Section 2 characterises the informative equilibrium in which the organisation motivates the agent's high-precision project selection by setting a threshold stricter than the screening-optimal threshold. In practice, promotion decisions also depend on the competition within the cohort; agents of high quality might be rejected due to a limited promotion quota and severe competition. In this extension, we analyse the case in which the organisation commits to the intensity of competition instead of a fixed threshold in the single-period setup. The analysis confirms that the informative mechanism of Section 2 is not an artefact of exogenous commitment: the agent's single-cutoff best response and the optimal informative commitment both survive when the threshold is determined endogenously by the cohort's realised performance.

We replace the single agent with a cohort of $N \geq 2$ ex ante identical agents (abilities are iid draws from the same ability distribution), and let the organisation commit to promoting exactly $k \in \{1, \ldots, N-1\}$ of them. The agents with the k highest estimated abilities are promoted.

11 Setup

There are $N \geq 2$ agents indexed by i. Each agent independently chooses a project precision $h_i \in [\underline{h}, \bar{h}]$ with $0 < \underline{h} < \bar{h}$. Conditional on h_i, agent i's posterior estimator of ability after one project is

$$\hat{\eta}_i \mid h_i \;\sim\; \mathcal{N}(B,\; \sigma_b(h_i)^2), \qquad \sigma_b(h) \;:=\; \sqrt{\tfrac{h}{h_\eta(h+h_\eta)}}, \tag{41}$$

exactly as in (9). The map $\sigma_b(h)$ is strictly increasing in h, so the agent's choice between $\underline{h}$ and $\bar{h}$ is equivalent to a choice between low and high posterior dispersion.

The organisation observes $(\hat{\eta}_1, \ldots, \hat{\eta}_N)$ and promotes the k agents with the largest estimates. Ties occur with probability zero because the estimators are normally distributed, so the tie-breaking rule is immaterial. We denote

$$\alpha \;:=\; \frac{k}{N} \tag{42}$$

for the promotion ratio.

12 The agent's problem

The game is Stackelberg: the organisation commits to k as leader, and the N agents play a simultaneous-move precision subgame as followers. We solve by backward induction, focusing on symmetric Nash equilibria of the agents' subgame — equivalently, on agent i's best response against a symmetric opponent project profile.

Suppose agent i chooses h' while all other agents choose a common h. Let

$$\rho \;:=\; \frac{\sigma_b(h')}{\sigma_b(h)}, \qquad T_i \;:=\; \frac{\hat{\eta}_i - B}{\sigma_b(h')}, \tag{43}$$

so $T_i \sim \mathcal{N}(0,1)$ is agent i's standardised posterior. The ratio ρ measures the deviator's posterior dispersion relative to the opponents', in standard-deviation rather than precision language. Conditional on $T_i = t$, each opponent's posterior independently follows a normal distribution with mean B and standard deviation $\sigma_b(h)$, so

$$\begin{aligned} \Pr(\hat{\eta}_i > \hat{\eta}_j \mid T_i = t) &= \Pr\big(\sigma_b(h)\, T_j < \sigma_b(h')\, t \mid T_i = t\big) && (44)\\ &= \Pr(T_j < \rho\, t \mid T_i = t) && (45)\\ &= \Phi(\rho\, t). && (46) \end{aligned}$$

The number of opponents whose estimates fall below agent i is therefore binomial, and agent i is promoted iff at least $N-k$ of them fall below. Averaging over T_i,

$$\pi(\rho;\, N, k) \;=\; \int_{-\infty}^{\infty} \phi(t)\, \Pr\Big(\mathrm{Binomial}\big(N-1,\, \Phi(\rho t)\big) \geq N-k\Big)\, dt. \tag{47}$$

Here $\pi(\rho; N, k)$ is agent i's ex ante promotion probability under the precision profile (h', h), which enters only through the relative-dispersion ratio ρ. Under a symmetric profile, $\rho = 1$ and all N posteriors are iid $\mathcal{N}(B, \sigma_b(h)^2)$, so $\pi(1; N, k) = k/N$.

Lemma 12.1 shows that the direction in which the ex ante promotion probability moves with a marginal increase in relative dispersion depends on whether the organisation promotes fewer than half the cohort.

Lemma 12.1 (Promotion-probability comparative static)**.** *For every $\rho > 0$ and every $k \in \{1, \dots, N-1\}$,*

$$\operatorname{sgn} \frac{\partial \pi(\rho;\, N, k)}{\partial \rho} \;=\; \operatorname{sgn}(N - 2k). \tag{48}$$

Proof. See Appendix C. □

The intuition is the same distributional logic that runs through Section 3. When weakly fewer than half the cohort is promoted, promotion requires reaching the upper tail of the rank distribution, and higher dispersion relative to the cohort makes the upper tail thicker, which is in the agent's favour. When more than half is promoted, the binding risk is falling into the lower tail, and greater dispersion stretches that tail against the agent.

The lemma delivers best responses of the same form as Lemma 3.1. Since σ_b is strictly monotone in h, π is strictly monotone in h_i for every opponent profile whenever $k \neq N/2$. The best response is therefore $h_i = \bar{h}$ when $k < N/2$ and $h_i = \underline{h}$ when $k > N/2$.

At $k = N/2$ the agent is indifferent across all h_i, and we adopt the same favourable tie-breaking convention there: the agent selects the highest-precision project among the indifferent precision values. Under this convention the symmetric equilibrium is unique:

$$h_i^*(k) \;=\; \begin{cases} \bar{h}, & k \le N/2, \\ \underline{h}, & k > N/2. \end{cases} \tag{49}$$

Competition therefore motivates ability revelation whenever the organisation commits to promoting no more than half the cohort.

13 The organisation's payoff

The organisation receives utility equal to the sum of the posterior estimates of each promoted agent. This setup captures the same trade-off as in (10): the organisation is rewarded by a better estimator and punished by a low promotion ratio.

At a symmetric equilibrium all agents choose the same $h^*(k)$ from (49), so the posteriors are iid $\mathcal{N}(B, \sigma_b(h^*(k))^2)$ and the top-k promoted posteriors correspond to the top-k order statistics of $(Z_1, \dots, Z_N) \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$. Writing

$$S(N,k) \;:=\; \mathbb{E}\left[\sum_{j=N-k+1}^{N} Z_{(j)}\right] \tag{50}$$

for the expected sum of the top k standard normal order statistics, the organisation's per-agent payoff is

$$V_N(k) \;=\; \frac{k}{N}\, B \;+\; \frac{\sigma_b(h^*(k))}{N}\, S(N,k). \tag{51}$$

This decomposes into the same quantity–quality trade-off as Section 4: k/N is the fraction promoted and $\sigma_b(h^*(k))\, S(N,k)/k$ is the expected posterior of a promoted agent. Similar to Section 2, we focus on the case $B > 0$.

Within the informative regime $k \le N/2$, we have $h^*(k) = \bar{h}$, $k \mapsto kB$ is linear, and $k \mapsto S(N,k)$ is strictly increasing (each increase in quota adds one more candidate from the top half of the cohort, i.e. agents with positive expected ability). Hence V_N is strictly increasing on the informative regime and the optimal informative quota is

$$k^*_{\text{informative}} \;=\; \lfloor N/2 \rfloor,$$

and is also the most generous quota in the regime. This is the quota analogue of the informative threshold with the least screening distortion $\eta^* = B$ in Section 2: both designs reach the boundary of the informative regime.

14 Interpretation

Section 2 commits to a deterministic threshold $\eta^* = B$, the most lenient threshold consistent with informative play. The quota replaces this exogenous commitment with a quantity commitment $k = \lfloor N/2 \rfloor$, the most generous quota consistent with informative play. Commitment is not eliminated; it is relocated from a value to a count.

The mechanism is identical. Both designs implement $h^* = \bar{h}$ through the same single-cutoff best response: although the quota replaces the deterministic threshold with the random binding cutoff $\hat{\eta}_{(N-k+1)}$, normal symmetry of the cohort's posteriors collapses the agent's comparison to whether the binding cutoff lies above the prior mean, with $\eta^* \gtrless B$ replaced by $k \lesseqgtr N/2$. Both designs also require the precision range $[\underline{h}, \bar{h}]$ to be wide enough that the quality premium $\sigma_b(\bar{h}) - \sigma_b(\underline{h})$ outweighs the quantity restriction informative play imposes — the threshold gives up screening against the ex-post optimum $\eta^* = 0$, the quota gives up promoting beyond $\lfloor N/2 \rfloor$. Where the two designs diverge is in the cutoff each effectively applies: the threshold cuts at the population median B, the quota at the sample median of the cohort. At finite N the sample median fluctuates around B, and this selection noise costs the organisation a small welfare loss relative to the threshold design.

As $N \to \infty$ the sample median concentrates at B, the quota's selection rule converges to the threshold's, and the per-agent payoffs $V_N(\lfloor N/2 \rfloor)$ and U_p^h coincide. Competition therefore implements the threshold-design outcome in the large-cohort limit, with finite-cohort welfare strictly below.

15 Discussion

The result relies on the concentrated density around the mean and the thin tails of the posterior distribution, so it could stand with more general distributions; we rely on normality for tractability of posterior estimators. The precision-parametrised project menu is highly restrictive, but it captures a feature of project outcomes in the career setting: when conducting a project, the agent is less likely to choose any posterior distribution she likes. She picks among feasible projects, and projects carry their own variance properties.

Normality does not save us from tractability issues in the multi-period setup. While we elicit the key mechanisms in particular the recovery–safety trade-off, the equilibrium in the multi-period setting, and how close the substitution between probation length and threshold strictness is, both require future work.

Two extensions sit close enough to the present model: a technical one and a structural one. The first is a fully multi-period equilibrium. The two-period analysis already opens onto a dynamic programming problem on the agent's side: at every stage of probation she decides how much precision to spend on the current project, weighing the chance of recovering from a poor early draw against the value of locking in a good one. The technical obstacle is that her payoff over precision profiles is relatively flat near the optimum: which requires a large number of grid points in computation. The backward induction resists the closed-form treatment that two periods admit. Solving it numerically would deliver the agent's true best response across the horizon, and let the joint design of strictness and duration be solved beyond the constant-precision bound used here.

The second is interim action by the organisation. In practice the organisation often has the chance to fire or promote an agent before the end of the probation period, especially when the agent achieves exceptional or disastrous early progress. A future extension might model this mechanism as additional thresholds on the interim posterior estimate, which would be expected to generate better ability revelation through higher precision — or, in the standard effort–wage context, through excessive effort.

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A Proofs

B Proofs for Section 6

Proof of Lemma 8.2. At $\eta^* = B$, $z^* = 0$, and (31) reduces to

$$\mathcal{P}(h_1; B) \;=\; \int_{-\infty}^{0} \Phi(r^H z)\varphi(z)\,dz \;+\; \int_{0}^{\infty} \Phi(r^L z)\varphi(z)\,dz. \tag{52}$$

Substituting $u = -z$ in the first integral and using $\varphi(-u) = \varphi(u)$, $\Phi(-x) = 1 - \Phi(x)$,

$$\int_{-\infty}^{0} \Phi(r^H z)\varphi(z)\,dz \;=\; \frac{1}{2} - \int_{0}^{\infty} \Phi(r^H u)\varphi(u)\,du, \tag{53}$$

so that

$$\mathcal{P}(h_1; B) \;=\; \frac{1}{2} + I(r^L) - I(r^H), \qquad I(a) \;:=\; \int_0^\infty \Phi(au)\,\varphi(u)\,du. \tag{54}$$

Evaluate $I(a)$ for $a > 0$. Let $X, Y \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$. Reading $\Phi(au) = \Pr(X \le au)$,

$$I(a) \;=\; \Pr\!\big(X \le aY,\; Y > 0\big). \tag{55}$$

Set $V := X - aY$, so $V \sim \mathcal{N}(0, 1 + a^2)$ and $\mathrm{Cov}(V, Y) = -a$. With $\tilde{V} := V/\sqrt{1 + a^2}$, the pair $(\tilde{V}, -Y)$ is bivariate standard normal with correlation $\rho = a/\sqrt{1 + a^2}$. Sheppard’s orthant formula Sheppard (1898)yields

$$\Pr\!\big(\tilde{V} \le 0,\; -Y \le 0\big) \;=\; \frac{1}{4} + \frac{1}{2\pi}\arcsin\!\big(a/\sqrt{1+a^2}\big) \;=\; \frac{1}{4} + \frac{1}{2\pi}\arctan(a), \tag{56}$$

where the last equality uses $\arcsin\!\big(a/\sqrt{1+a^2}\big) = \arctan(a)$. Substituting into (54) and cancelling the 1/4 terms gives (32).

The expressions in (33) follow from $\sigma_1^2 = h_1/[h_\eta(h_\eta + h_1)]$, $(\sigma_2^L)^2 = \underline{h}/[H_1(H_1 + \underline{h})]$, $(\sigma_2^H)^2 = \bar{h}/[H_1(H_1 + \bar{h})]$ with $H_1 = h_\eta + h_1$, and $r^j = \sigma_1/\sigma_2^j$.

Existence and uniqueness of an interior maximiser on $(0,\infty)$ are established by Proposition B.1 below. □

Proposition B.1 (Interior best response at $T = 2$, $\eta^* = B$)**.** *The function $\mathcal{P}(h_1; B)$ defined in (32)–(33) is strictly quasi-concave on $h_1 \in (0,\infty)$, and admits a unique maximiser $h_1^* \in (0,\infty)$ characterised by the first-order condition*

$$\frac{(r^L)'(h_1)}{1 + (r^L(h_1))^2} \;=\; \frac{(r^H)'(h_1)}{1 + (r^H(h_1))^2}. \tag{57}$$

Proof. Set $D(h_1) := \arctan(r^L(h_1)) - \arctan(r^H(h_1))$, so $\mathcal{P} = \frac{1}{2} + \frac{1}{2\pi}D$. Since affine transformations preserve quasi-concavity, it suffices to prove D strictly quasi-concave.

Normalisation. Let $x := h_1/h_\eta$, $a := \underline{h}/h_\eta$, $b := \bar{h}/h_\eta$, with $0 < a < b$. From (33),

$$r^L = \sqrt{\tfrac{x(x+1+a)}{a}}, \qquad r^H = \sqrt{\tfrac{x(x+1+b)}{b}}. \tag{58}$$

Define $u_c(x) := \sqrt{x(x+1+c)/c}$ for $c \in \{a,b\}$; then $D(x) = \arctan u_a(x) - \arctan u_b(x)$, and quasi-concavity in h_1 is equivalent to quasi-concavity in x.

Derivative. Direct computation gives

$$u_c'(x) = \frac{2x+1+c}{2\sqrt{c}\,\sqrt{x}\,\sqrt{x+1+c}}, \qquad 1 + u_c(x)^2 = \frac{(x+c)(x+1)}{c}, \tag{59}$$

the second using the factorisation $c + x(x+1+c) = (x+c)(x+1)$. Hence

$$\frac{u_c'(x)}{1+u_c(x)^2} \;=\; \frac{\sqrt{c}\,(2x+1+c)}{2\sqrt{x}\,(x+c)(x+1)\,\sqrt{x+1+c}}, \tag{60}$$

and $D'(x)$ is the difference of (60) at $c = a$ and $c = b$.

First-order condition. Both terms in (60) are strictly positive on $x > 0$, so $D'(x) = 0$ is equivalent, after squaring, to

$$\frac{a(2x+1+a)^2}{(x+a)^2(x+1+a)} \;=\; \frac{b(2x+1+b)^2}{(x+b)^2(x+1+b)}. \tag{61}$$

Cross-multiplying, define

$$F(x) \;:=\; a(2x+1+a)^2(x+b)^2(x+1+b) - b(2x+1+b)^2(x+a)^2(x+1+a). \tag{62}$$

Direct expansion factorises F as

$$F(x) \;=\; (a-b)(x+1)\,Q(x), \tag{63}$$

where

$$\begin{aligned} Q(x) \;&= 4x^4 + 4(a+b+1)x^3 + (a^2+ab+2a+b^2+2b+1)x^2 \\ &\quad - ab(a+b+2)x - ab(ab+a+b+1). \end{aligned} \tag{64}$$

Since $a < b$ and $x + 1 > 0$ on $x > 0$,

$$D'(x) = 0 \iff Q(x) = 0. \tag{65}$$

At most one critical point. The coefficients of Q in (64) have sign pattern $(+,+,+,-,-)$: exactly one sign change. By Descartes' rule of signs, Q has at most one positive root, so D' has at most one zero on $(0,\infty)$.

Sign of D' at the boundary. As $x \to 0^+$, (60) gives

$$D'(x) \ \sim\ \frac{1}{2\sqrt{x}}\Big(\sqrt{\tfrac{1+a}{a}} - \sqrt{\tfrac{1+b}{b}}\Big). \tag{66}$$

Since $t \mapsto (1+t)/t$ is strictly decreasing on $(0,\infty)$, $a < b$ implies the bracket is positive; hence $D'(x) > 0$ for x sufficiently small. As $x \to \infty$, (60) expands to

$$D'(x) \ \sim\ \frac{\sqrt{a}-\sqrt{b}}{x^2} \ <\ 0. \tag{67}$$

By continuity, D' has at least one zero on $(0,\infty)$, which by the previous step is unique. Call it $\tilde{x}$. Then $D' > 0$ on $(0,\tilde{x})$ and $D' < 0$ on $(\tilde{x},\infty)$, so D is strictly increasing then strictly decreasing, hence strictly quasi-concave. The unique maximiser of $\mathcal{P}(\cdot;B)$ on $(0,\infty)$ is $h_1^* = h_\eta\tilde{x}$, and the FOC (57) is the equality of the two instances of (60). □

Proof of Proposition 9.1. We proceed in three steps.

Step 1. Strict concavity of $\sigma_T(h)$ in T. Write $H := Th$. Then

$$\sigma^2(H) = \frac{H}{h_\eta(h_\eta+H)},$$

and direct differentiation gives

$$\frac{d\sigma^2}{dH} = \frac{1}{(h_\eta+H)^2} > 0, \qquad \frac{d^2\sigma^2}{dH^2} = \frac{-2}{(h_\eta+H)^3} < 0,$$

with

$$\sigma(H) = \left(\frac{H}{h_\eta(h_\eta+H)}\right)^{1/2}.$$

$$\sigma'(H) = \frac{\sqrt{h_\eta}}{2\,H^{1/2}(h_\eta+H)^{3/2}}, \qquad \sigma''(H) = -\frac{\sqrt{h_\eta}\,(h_\eta+4H)}{4\,H^{3/2}(h_\eta+H)^{5/2}} < 0 \quad \text{for all } H>0.$$

So σ^2 and σ are strictly concave and strictly increasing in H. The composition $\sigma_T(h) = \sigma(H(T))$ is strictly concave in T.

Step 2. Informative regime. The gross payoff satisfies $U_I(T) + cT = B/2 + \phi(0)\sigma_T(\bar{h})$, a strictly increasing affine transformation of $\sigma_T(\bar{h})$. By Step 1 this is strictly concave in T unconditionally. Since $\sigma_T(\bar{h}) \to 1/\sqrt{h_\eta}$ while $c(T-1) \to \infty$, we have $U_I(T) \to -\infty$, and a unique finite maximiser exists.

Step 3. Uninformative regime under condition (SC). The gross payoff is $G_{\underline{h}}(T) := g(\sigma_T(\underline{h}))$, where $g(\sigma) := B\Phi(B/\sigma) + \sigma\phi(B/\sigma)$. Computing derivatives:

$$g'(\sigma) = \phi\Big(\frac{B}{\sigma}\Big), \qquad g''(\sigma) = \frac{B^2}{\sigma^3}\,\phi\Big(\frac{B}{\sigma}\Big) > 0,$$

so g is strictly convex in σ. By the chain rule,

$$G''_{\underline{h}}(T) = g''(\sigma_T)\,(\sigma'_T)^2 + g'(\sigma_T)\,\sigma''_T = \phi\Big(\frac{B}{\sigma_T}\Big)\left[\frac{B^2}{\sigma_T^3}(\sigma'_T)^2 + \sigma''_T\right],$$

where $\sigma_T := \sigma_T(\underline{h})$. Since $\phi(B/\sigma_T) > 0$, the condition $G''_{\underline{h}}(T) < 0$ reduces to

$$B^2 < \frac{-\sigma''_T\,\sigma_T^3}{(\sigma'_T)^2}.$$

We compute the right-hand side. Denote $y := T\underline{h}/h_\eta$, so that $\sigma_T = h_\eta^{-1/2}(y/(1+y))^{1/2}$. Direct differentiation yields

$$\sigma_T' = \frac{\underline{h}/h_\eta}{2\,h_\eta^{1/2}\,y^{1/2}(1+y)^{3/2}}, \qquad \sigma_T'' = -\frac{(\underline{h}/h_\eta)^2(1+4y)}{4\,h_\eta^{1/2}\,y^{3/2}(1+y)^{5/2}},$$

from which

$$\frac{-\sigma_T''\,\sigma_T^3}{(\sigma_T')^2} = \frac{y(1+4y)}{h_\eta(1+y)} =: \frac{R(y)}{h_\eta}.$$

Since $R'(y) = (1+8y+4y^2)/(1+y)^2 > 0$, R is strictly increasing. At $T=1$, $y = \underline{h}/h_\eta$, so it is sufficient to require

$$B^2 h_\eta < R\Big(\frac{\underline{h}}{h_\eta}\Big) = \frac{(\underline{h}/h_\eta)(1+4\,\underline{h}/h_\eta)}{1+\underline{h}/h_\eta},$$

which is condition (SC). Under (SC), $G''_{\underline{h}}(T) < 0$ for all $T \geq 1$, so $U_U(T)$ is strictly concave. Since $\sigma_T(\underline{h})$ and hence $g(\sigma_T(\underline{h}))$ is bounded above as $T \to \infty$ while $c(T-1) \to \infty$, the same divergence argument as in Step 2 gives a unique finite maximiser. □

Proof of Proposition 9.2. By Proposition 9.1, U_I is strictly concave in T, so $U_I'(T) = \phi(0)\sigma_T'(\bar{h}) - c$ is strictly decreasing. Strict concavity then implies $T_I^* \geq 2 \iff U_I'(2) \geq 0 \iff c \leq \phi(0)\sigma_2'(\bar{h})$: in the forward direction, if $T_I^* \geq 2$ then either it is interior with $U_I'(T_I^*) = 0$, so $U_I'(2) \geq U_I'(T_I^*) = 0$ by monotonicity, or T_I^* is at the boundary $T = 2$ of an infeasible interior optimum, again with $U_I'(2) \geq 0$; in the reverse direction, $U_I'(2) \geq 0$ together with the decreasing U_I' implies U_I is non-decreasing on $[1,2]$, so the unique maximiser lies in $[2,\infty)$. The U-regime argument is identical under condition (SC), using $U_U'(T) = \phi(B/\sigma_T(\underline{h}))\,\sigma_T'(\underline{h}) - c$. □

Proof of Lemma 9.3. **Step 1. σ_H as a function of σ_L.** From $\sigma_T^2(h) = Th/[h_\eta(h_\eta + Th)]$, inverting for T gives $T = h_\eta^2\sigma_L^2/[\underline{h}\,(1-h_\eta\sigma_L^2)]$. Substituting into $\sigma_H^2 = T\bar{h}/[h_\eta(h_\eta + T\bar{h})]$ and simplifying:

$$\sigma_H^2 = \psi(\sigma_L)^2 = \frac{r\,\sigma_L^2}{1+a\,\sigma_L^2}, \qquad r := \frac{\bar{h}}{\underline{h}}, \quad a := (r-1)h_\eta.$$

Equivalently $\psi(\sigma_L) = \sigma_L\sqrt{r/(1+a\sigma_L^2)}$. Differentiating once and twice:

$$\psi'(\sigma_L) = \frac{\sqrt{r}}{(1+a\,\sigma_L^2)^{3/2}}, \qquad \psi''(\sigma_L) = -\frac{3a\sqrt{r}\,\sigma_L}{(1+a\,\sigma_L^2)^{5/2}}. \tag{68}$$

Since $\bar{h} > \underline{h}$, $a = (r-1)h_\eta > 0$, so $\psi' > 0$ and $\psi'' < 0$ on $\sigma_L > 0$.

Step 2. First and second derivatives of $\tilde{F}$. Writing $F_T = B/2 + \phi(0)\,\sigma_H - g(\sigma_L)$ with $g(\sigma_L) := B\,\Phi(B/\sigma_L) + \sigma_L\,\phi(B/\sigma_L)$, the truncated-normal identity gives $g'(\sigma_L) = \phi(B/\sigma_L)$, so

$$\tilde{F}'(\sigma_L) = \phi(0)\,\psi'(\sigma_L) - \phi\Big(\frac{B}{\sigma_L}\Big).$$

Differentiating again, using $\phi'(x) = -x\phi(x)$ to obtain $\frac{d}{d\sigma_L}\phi(B/\sigma_L) = (B^2/\sigma_L^3)\,\phi(B/\sigma_L)$:

$$\tilde{F}''(\sigma_L) = \phi(0)\,\psi''(\sigma_L) - \frac{B^2}{\sigma_L^3}\,\phi\Big(\frac{B}{\sigma_L}\Big).$$

Step 3. $\tilde{F}$ is strictly concave in σ_L. Both terms of $\tilde{F}''(\sigma_L)$ are strictly negative: the first by $\psi'' < 0$ (Step 1), the second because $B^2, \sigma_L^3, \phi(B/\sigma_L) > 0$. Hence $\tilde{F}'' < 0$ and $\tilde{F}$ is strictly concave in σ_L.

Step 4. Quasi-concavity in T. By Step 1 of the proof of Proposition 9.1, $\sigma_L(T) = \sigma_T(\underline{h})$ is strictly increasing in T. Since $\tilde{F}$ is strictly concave in σ_L and $\sigma_L(T)$ is strictly increasing, every upper level set $\{T > 0 : F_T \geq k\} = \{T > 0 : \tilde{F}(\sigma_L(T)) \geq k\}$ is an interval. Hence F_T is strictly quasi-concave in T. □

C Proof of Lemma 12.1

The argument has three ingredients: a binomial-tail derivative identity, differentiation under the integral sign, and a symmetrisation in t.

Step 1: Binomial-tail derivative identity. For integers $n \geq 1$, $m \in \{1, \ldots, n\}$, and $u \in (0,1)$,

$$\partial_u \Pr\bigl(\text{Binomial}(n,u) \geq m\bigr) \;=\; n\binom{n-1}{m-1} u^{m-1}(1-u)^{n-m}. \tag{69}$$

Differentiating $\sum_{j=m}^{n} \binom{n}{j} u^j (1-u)^{n-j}$ termwise and using $j\binom{n}{j} = n\binom{n-1}{j-1}$ and $(n-j)\binom{n}{j} = n\binom{n-1}{j}$ produces two sums that, after reindexing the first by $i = j-1$, share a common range except for a single boundary term at $i = m-1$. The cancellation leaves exactly the right-hand side of (69).

Step 2: Differentiation under the integral. Setting $n = N-1$, $m = N-k$, $u = \Phi(\rho t)$ in (69) and applying the chain rule, the integrand of (47) differentiates to a positive multiple of $t\,\phi(t)\,\phi(\rho t)\,\Phi(\rho t)^{N-k-1}(1-\Phi(\rho t))^{k-1}$. This is dominated by $C\,|t|\,\phi(t)$ with $C = (N-1)\binom{N-2}{N-k-1}/\sqrt{2\pi}$, since $\phi(\rho t) \leq 1/\sqrt{2\pi}$ and both Φ-factors lie in $[0,1]$; the dominator is integrable on $\mathbb{R}$, so differentiation under the integral is justified for all $\rho > 0$, and

$$\partial_\rho \pi(\rho;\, N, k) \;\propto\; \int_{-\infty}^{\infty} t\,\phi(t)\,\phi(\rho t)\,\Phi(\rho t)^{N-k-1}\bigl(1-\Phi(\rho t)\bigr)^{k-1}\,dt, \tag{70}$$

with strictly positive constant of proportionality $(N-1)\binom{N-2}{N-k-1}$.

Step 3: Symmetrisation. Split the integral in (70) at $t = 0$ and substitute $t \mapsto -t$ on the negative half. Using $\phi(-x) = \phi(x)$ and $\Phi(-x) = 1 - \Phi(x)$,

$$\partial_\rho \pi(\rho;\, N, k) \;\propto\; \int_0^{\infty} t\,\phi(t)\,\phi(\rho t)\,\Bigl[\Phi(\rho t)^{N-k-1}(1-\Phi(\rho t))^{k-1} \;-\; (1-\Phi(\rho t))^{N-k-1}\Phi(\rho t)^{k-1}\Bigr]\,dt. \tag{71}$$

For $\rho > 0$, $u := \Phi(\rho t) \in (1/2, 1)$ on $(0, \infty)$. The bracket factors as

$$u^{k-1}(1-u)^{k-1}\bigl[u^{N-2k} - (1-u)^{N-2k}\bigr],$$

where the exponents u^{N-2k} and $(1-u)^{N-2k}$ are well defined for any integer $N-2k$ on $u \in (1/2, 1)$. The map $x \mapsto x^{N-2k}$ on $(0,1)$ is strictly increasing if $N - 2k > 0$, constant if $N - 2k = 0$, and strictly decreasing if $N - 2k < 0$. Combined with $u > 1 - u > 0$,

$$\operatorname{sgn}\bigl[u^{N-2k} - (1-u)^{N-2k}\bigr] \;=\; \operatorname{sgn}(N-2k).$$

The weight $t\,\phi(t)\,\phi(\rho t)\,u^{k-1}(1-u)^{k-1}$ is strictly positive on $(0,\infty)$, so the bracket's sign is preserved under integration. When $N \neq 2k$ the integrand has a constant non-zero sign on a non-degenerate interval, so $\partial_\rho \pi(\rho; N, k)$ is strictly of sign $N - 2k$. When $N = 2k$ the bracket vanishes identically and $\partial_\rho \pi \equiv 0$. □

D Sicomparative-statics

E Comparative Statics with Respect to Benchmark Precision

We examine how the critical dispersion $x^*(s)$ varies with the benchmark precision s. By implicit differentiation of

$$F(x^*(s), s; B, h_\eta) = 0, \tag{72}$$

we obtain

$$\frac{dx^*}{ds} = -\frac{\partial F/\partial s}{\partial F/\partial x}. \tag{73}$$

Since $\partial F/\partial x > 0$ under the existence condition, the sign of dx^*/ds depends on $\partial F/\partial s$.

Derivative with respect to s

Differentiating F with respect to s yields

$$\frac{\partial F}{\partial s} = \sigma_b'(\bar{h})\phi(0) - \sigma_b'(\underline{h})\phi\left(\frac{B}{\sigma_b(\underline{h})}\right), \tag{74}$$

where

$$\sigma_b'(h_p) = \frac{\partial \sigma_b(h_p)}{\partial h_p} = \frac{1}{2}\frac{\sqrt{h_\eta}}{\sqrt{h_p}(h_p + h_\eta)^{3/2}} > 0. \tag{75}$$

Limiting behaviour

At the extended limit $B \to \frac{2}{\sqrt{h_\eta}}\phi(0)$, the organisation exploits the maximum information advantage in the informative equilibrium, that is, $s \to \infty, x^* \to s^-$, equivalently, $\bar{h} \to \infty, \underline{h} \to 0$.

The partial derivative becomes $\lim_{\bar{h}\to\infty} \sigma_b'(\bar{h})\phi(0) - \lim_{\underline{h}\to 0} \sigma_b'(\underline{h})\phi\left(\frac{B}{\sigma_b(\underline{h})}\right)$, the second limit vanishes, since the exponential decay of the standard Normal pdf dominates the square root, the $\phi\left(\frac{B}{\sigma_b(\underline{h})}\right)$ decay to zero faster than $\sigma_b'(\underline{h})$ to infinity, and the first term takes the limit to zero. For a moderate B and s is extremely close to the threshold in which the second term diminishes for $\underline{h} \to 0$, and the first term becomes $\sigma_b'(2s)\phi(0)$ which makes the entire term strictly positive. We show this is a plausible state through a numerical example and the rarity through a simulation.

We examined the case where $h_\eta = 1, B = 0.4$, the inequality 20 now could be written as

$$\begin{aligned} s &> \tfrac{B^2 h_\eta^2}{2(4\phi(0)^2 - B^2 h_\eta)} && (76)\\ &= \frac{0.4^2}{8\phi(0)^2 - 2 \times 0.4^2} && (77)\\ &\approx 0.168 && (78)\end{aligned}$$

At $s = 0.196, x = 0.15$ we have $F(x = 0.15, s = 0.196, B = 0.4, h_\eta = 1) = -6.596485e - 06$, and the partial derivative $\frac{\partial F}{\partial s} \approx 0.075$ is positive. The result implying that the increase in the benchmark project precision s reduces the critical project dispersion x^*. As the first order partial derivative is weakly negative at two ends of the feasible range of B, we infer that this positive impact is largest at a moderate B, and its rarity is validated by Figure 4 where $B = 0.4 \approx \frac{1}{2}\frac{2}{\sqrt{h_\eta}}\phi(0), h_\eta = 1$.

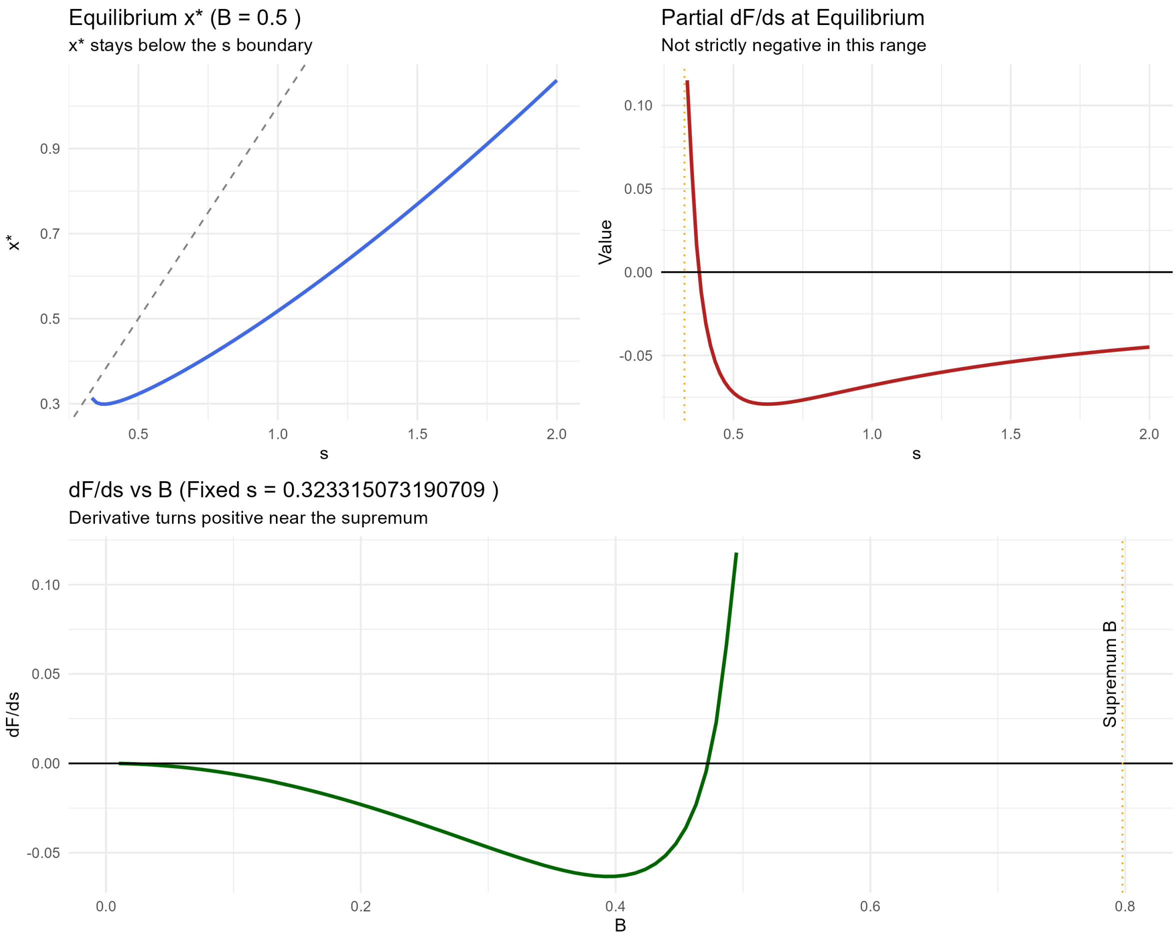

Figure 4: A numerical example with B = 0.4, $h_\eta = 1$